SEAT NO. CE-24009

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

FIRST YEAR (SOFTWARE/ ECONOMIC & FINANCE)
SPRING SEMESTER EXAMINATIONS 2025
Batch 2024

Dated:14-06-2025

Time: 3 Hours

Max.Marks:60

ACADEMIC READING & WRITING – EA-115

Note: Attempt all questions.

Read the passage below and answer the given questions:

The Future of Renewable Energy
David M. Thompson

Renewable energy has become a focal point in the quest for a sustainable and environmentally friendly future. It sparks various visions and expectations in the minds of different individuals. Some foresee a world powered entirely by clean, renewable sources like solar, wind, hydro, and geothermal energy, reducing our reliance on fossil fuels and mitigating the impacts of climate change. Others imagine vast solar farms and wind turbines dotting the landscape, harnessing the natural power of the elements to generate electricity for homes and industries. Some envision innovative energy storage solutions, such as advanced batteries and hydrogen fuel cells, facilitating the widespread adoption of renewable energy and enabling a more resilient power grid. Yet, there are concerns about the challenges of transitioning from conventional energy sources to renewables, including grid integration, intermittency, and the need for significant infrastructural changes.

As of today, renewable energy technologies have made impressive strides, but their full potential is yet to be realized. Solar and wind power have experienced exponential growth, becoming increasingly competitive with traditional fossil fuel-based energy sources. Countries and regions are setting ambitious targets for renewable energy adoption, aiming to achieve a carbon-neutral future.

The future of renewable energy holds exciting prospects for both the environment and the economy. Advancements in photovoltaic cells and wind turbine designs could lead to even greater energy efficiency and cost-effectiveness. Breakthroughs in energy storage technologies will address the challenge of intermittency, enabling renewable energy to supply a more stable and reliable power grid. Additionally, the integration of artificial intelligence and smart grid technologies could optimize the management and distribution of renewable energy resources.

Renewable energy's impact on society will be profound. The widespread adoption of clean energy will reduce greenhouse gas emissions, contributing to global efforts to combat climate change and improve air quality. Moreover, the shift towards renewables will create new job opportunities and drive economic growth in the renewable energy sector.

However, there are obstacles to overcome. Developing countries may face challenges in adopting renewable energy due to initial infrastructure costs and technological barriers. Additionally, the existing energy infrastructure, dominated by fossil fuels, presents complexities in transitioning to a renewable-centric energy system.

The future of renewable energy hinges on our collective commitment to innovation, investment, and international collaboration. Policymakers must create supportive regulatory frameworks and incentives to accelerate the transition to clean energy. Research and development efforts should focus on increasing the efficiency and scalability of renewable technologies.

In conclusion, the future of renewable energy is full of potential, promising a cleaner, greener, and more sustainable world. By embracing renewable energy solutions and addressing the challenges ahead, we can pave the way for a brighter future for generations to come, where our energy needs are met with minimal impact on the planet's ecosystem.

Q. 01: a. Read the above passage and answer the following questions:
[CLO-1] (Marks 10)

1. According to the text, what are some of the different visions and expectations people have regarding the future of renewable energy?
2. What are the main renewable energy sources mentioned in the text, and how do they contribute to reducing our reliance on fossil fuels?
3. Discuss the environmental and economic benefits of transitioning from conventional energy sources to renewable energy. Support your arguments with examples from the text.
4. Evaluate the potential of photovoltaic cells and wind turbine designs in achieving greater energy efficiency and cost-effectiveness for renewable energy systems. Provide evidence from the text and other relevant sources.

b. Write the contextual meanings of the following words from the text.
[CLO-1] (Marks 10)

1. Harnessing
2. Visions
3. Fossil fuels
4. Photovoltaic
5. Intermittency
6. Mitigating
7. Breakthroughs
8. Resilient
9. Embracing
10 Emissions

Q. 02: Read the above passage and write a concise summary that captures the main points and key information. Write a summary of the passage in no more than 100 words. Ensure your summary captures the main points about Renewable Energy.
[CLO-03] (Marks 10)

Q. 03. You are tasked with writing a research paper on the impact of renewable energy on economic growth. Properly cite the following sources using the IEEE/APA referencing style:
[CLO-03] (Marks 10)

1. A journal article titled "Renewable Energy and Its Contributions to Economic Growth" by Jessica Parker and Mark Davis, published in the Journal of Energy Economics, vol. 30, no. 2, pp. 78-91, 2019.
2. An online report titled "The Economic Implications of Renewable Energy Adoption" by the International Energy Agency (IEA), published in 2020.
3. A book titled "The Role of Renewable Energy in Fostering Economic Development" written by Laura Adams, published by Routledge in 2018.
4. A conference paper titled "Renewable Energy Policies and Their Impact on Employment in Different Sectors" by Sarah Brown, presented at the Annual Conference on Sustainable Development, 2021.
5. A webpage from the World Bank website, providing statistics on the economic benefits of renewable energy adoption in various countries for the year 2021.

Q. 04: To what extent do you agree or disagree with the text where it says "Advancements in photovoltaic cells and wind turbine designs could lead to even greater energy efficiency and cost-effectiveness in renewable energy systems." **Justify your answer with examples from your *surroundings*. (200-250 words)** **[CLO-2] (Marks 10)**

Q. 05: The bar chart below shows the average rainfall for Australia by month for last year. The line shows the average rainfall for Australia by month for the last 40 years. Summarize the information by selecting and reporting the main features, and make comparisons where relevant. You should write at least 150 words. [CLO-2] [Marks 10]

Average Rainfall for Australia for Last Year by Month